Asymptotic Methods

Part II

Recorded from the lecturer in 2026-2027.

Pre-requisites

- Complex methods / analysis.
- Refer to Section II in Stuart.

1. Intro & Motivation

- Most real problems don't have closed form solutions, or if they do, they are unworkable or difficult to interpret.
- Numerical approximations can be useful, but they can be expensive, they may have bad numerical properties, or solutions may be inscrutable.
- Taylor series, etc. can be limited in applicability.
- Asymptotic methods emphasise usefulness, interpretability as well as accuracy.

Examples

e.g. 1)

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\, dt, \qquad \mathrm{Re}(z) > 0.$$

For n an integer,

$$\Gamma(n+1) = n! \sim \sqrt{2\pi}\, n^{n+\frac{1}{2}} e^{-n} \left(1 + \frac{1}{12n} + \frac{1}{288n^2} + \cdots\right) \qquad \text{as } n \to \infty.$$

e.g. 2)

$$x^2 y'' + xy' - x^2 y = 0.$$

Exact series solution:

$$I_0(x) = \sum_{n=0}^{\infty} \frac{x^{2n}}{2^{2n}(n!)^2}.$$

Liouville–Green:

$$\implies I_0(x) \sim (2\pi x)^{-1/2} e^x \qquad \text{as } x \to \infty.$$

e.g. 3)

$$y'' + k^2 e^{-x^2/2} y = 0,$$

with

$$y \to y_0 \qquad \text{as } x \to \infty,$$

$$y \to \pm y_0 \qquad \text{as } x \to -\infty.$$

$$k = ?, \qquad k = k_0, k_1, k_2, \ldots$$

$$\implies k_n \sim \frac{\sqrt{\pi}}{2}\left(n + \frac{1}{2}\right) \qquad \text{as } n \to \infty.$$

WKBJ.

- Convergent methods typically give arbitrarily accurate approximations with more & more terms/iterations.
- Asymptotic methods give arbitrarily accurate approximations as some variable approaches a limit. Thus they always come with a qualifier, e.g. $z \to 0$ or $n \to \infty$.

2. Asymptotic Expansions

2.1 Definitions & Notation

(a) Asymptotic Equality/Equivalence

For 2 functions $f(x)$ and $g(x)$, we have

$$f(x) \sim g(x) \qquad \text{about } x = x_0$$

if

$$\boxed{\lim_{x \to x_0} \frac{f(x)}{g(x)} = 1.}$$

Remarks:

- Usually we want to approximate f with g.
- We can write
$$f(x) = \underbrace{g(x)}_{\text{dominant}} + \underbrace{[f(x) - g(x)]}_{\text{subdominant}}.$$
- Note $\sim$ is an equivalence relation (that is reflexive, transitive, symmetric).
- It respects products:
$$f_1 \sim g_1, \qquad f_2 \sim g_2$$
$$\implies f_1 f_2 \sim g_1 g_2.$$
- It is linear:
$$\alpha f_1 + \beta f_2 \sim \alpha g_1 + \beta g_2,$$
where α, β are constants.
- Definition has problems when dealing with oscillatory functions that go through 0 as $x \to x_0$.

(b) "Little o"

Suppose f & g are real functions.

We write

$$\boxed{f(x) = o(g(x)) \qquad \text{as } x \to x_0}$$

if

$$\boxed{\lim_{x \to x_0} \frac{f(x)}{g(x)} = 0.}$$

Remark:

$$f(x) = o(g(x))$$

can also be written as

$$f(x) \ll g(x) \qquad \text{as } x \to x_0.$$

- Can be framed in inequalities, e.g. $x_0 = \infty$ and let $g > 0$.
  Then $f = o(g)$ as $x \to \infty$ is equivalent to

$$\forall \varepsilon > 0, \quad \exists a \quad \text{s.t. when } x > a,$$

$$\boxed{|f| \leq \varepsilon g.}$$

(c) "Big O"

We write

$$f(x) = O(g(x)) \qquad \text{as } x \to x_0$$

for $g > 0$, if

$$\exists C > 0 \quad \text{s.t.}$$

$$\boxed{|f(x)| \leq C g(x)}$$

for x sufficiently close to x_0.

Warning:

In some areas of physics people use big-O and $\sim$ interchangeably.

Little Exercises: Show That

(a)

$$f(x) \sim g(x) \iff f - g = o(g) \qquad \text{as } x \to x_0.$$

(b)

$$\sin x \ll x^2 \qquad \text{as } x \to \infty.$$

(c)

$$12x^5 + 6x^3 + x = O(x^5) \qquad \text{as } x \to \infty.$$

(d)

$$e^{ax} \ll e^{bx} \qquad \text{as } x \to \infty,$$

if $b > a > 0$.

(e)

$$e^x \gg x^N \qquad \text{as } x \to \infty,$$

$$\forall N \in \mathbb{Z}^+.$$

2.2 Asymptotic Expansions

- Have unworkable/unknown $f(x)$.
- Want to know its behaviour in a limit.
- Want to find a simpler/known $g(x)$ so that

$$f \sim g.$$

- Also want to break up g into a hierarchy of approximations, s.t.

$$g(x) = \sum_{k=0}^{N} a_k \phi_k(x),$$

where a_k are coefficients, ϕ_k are known & workable, and where adding more terms gives a better approximation that can be estimated.

(a) Asymptotic Sequence

A sequence of functions

$$\{\phi_0(x), \phi_1(x), \ldots\}$$

is an "asymptotic sequence" as $x \to x_0$ if

$$\boxed{\phi_{k+1}(x) \ll \phi_k(x) \qquad \text{as } x \to x_0, \quad \forall k.}$$

e.g. 1)

$$\phi_k = (x-a)^k \qquad \text{around } x = a.$$

Check:

$$\frac{\phi_{k+1}}{\phi_k} = x - a \to 0 \qquad \text{as } x \to a. \quad \checkmark$$

e.g. 2)

$$\phi_k = e^{-kx} x^k \qquad \text{as } x \to \infty \text{ or } x \to -\infty.$$

Check:

$$\frac{\phi_{k+1}}{\phi_k} = e^{-x} x.$$

$$e^{-x} x \to 0 \quad \text{as } x \to \infty \qquad \checkmark$$
$$e^{-x} x \to \infty \quad \text{as } x \to -\infty \qquad \times$$

e.g. 3)

$$\phi_k = \cos(nx) \qquad \text{as } x \to \infty.$$

$\times$

e.g. 4)

$$\{|x|, \log|x|, \log|\log|x||, \ldots\} \qquad \text{as } x \to \infty.$$

?

(b) Asymptotic Expansion

- Suppose $f(x)$ is a function we want to understand near $x = x_0$.
- Let $\{\phi_k\}_{k=0}^{\infty}$ be an asymptotic sequence.

Then, the finite sum

$$\sum_{k=0}^{N} a_k \phi_k(x)$$

for some coefficients, is an (N-term) asymptotic approximation of $f(x)$ as $x \to x_0$ if:

$$\boxed{f(x) = \sum_{k=0}^{N} a_k \phi_k(x) + o(\phi_N(x))} \qquad \text{as } x \to x_0.$$

Alternatively, we can write the above as

$$\boxed{\frac{f(x) - \sum_{k=0}^{N} a_k \phi_k(x)}{\phi_N(x)} \longrightarrow 0} \qquad \text{as } x \to x_0.$$

End of Lecture 1

(c) Asymptotic expansions/series

If (b) holds for *all* N, then this is called an "asymptotic series or expansion", and we write

$$f(x) \sim \sum_{k=0}^{\infty} a_k \phi_k(x) \qquad \text{as } x \to x_0.$$

Attention: No claim of convergence is made. It is just notation. We never fix x and then take $N \to \infty$.

Instead, we have an infinite number of approximations (of different N), and $x \to x_0$ is taken in each separately!

(d) Properties of asymptotic expansions and remarks

(I) Mathematical operations

Suppose $\{\phi_k(x)\}$ is an asymptotic sequence as $x \to x_0$, and let

$$f \sim \sum_{k=0}^{\infty} a_k \phi_k, \qquad g \sim \sum_{k=0}^{\infty} b_k \phi_k.$$

Linearity

$$\alpha f + \beta g \sim \sum_{k=0}^{\infty} (\alpha a_k + \beta b_k) \phi_k \qquad \text{as } x \to x_0.$$

Multiplication

If $\phi_k = (x - x_0)^k$, then

$$fg \sim \sum_{k=0}^{\infty} c_k \phi_k, \qquad c_k = \sum_{j=0}^{k} a_j b_{k-j}.$$

Division

See 3.8 BO.

Integration

Term-by-term integration is usually okay.

For example, if $\phi_k = (x - x_0)^k$, then

$$\int_{x_0}^{x} f(t)\,dt \sim \sum_{k=0}^{\infty} \frac{a_k (x - x_0)^{k+1}}{k+1} \qquad \text{as } x \to x_0.$$

Exercise: Use the definitions to prove this.

Differentiation

Term-by-term differentiation is not always okay.

To do this, you need $f'(x)$ to be continuous and to have an asymptotic series, i.e.

$$f'(x) \sim \sum_{k=0}^{\infty} c_k \phi_k(x).$$

(II) Different asymptotic representations

Suppose $\{\phi_k\}$ and $\{\psi_k\}$ are different asymptotic sequences as $x \to x_0$. Also,

$$f \sim \sum_{k=0}^{\infty} a_k \phi_k, \qquad f \sim \sum_{k=0}^{\infty} b_k \psi_k \qquad \text{as } x \to x_0.$$

Then a_k and b_k are not the same.

For example,

$$f(x) = 1 + \frac{1}{x}.$$

Consider $\phi_k = x^{-k}$ as $x \to \infty$. Then

$$f(x) \sim \sum_{k=0}^{\infty} a_k \phi_k, \qquad a_0 = 1, \quad a_1 = 1, \quad a_n = 0 \quad (n > 1).$$

Now consider

$$\psi_k = (1 + x)^{-k}.$$

Then

$$\begin{aligned} f = 1 + \frac{1}{x} &= \frac{1}{1 - (x+1)^{-1}} \\ &= 1 + \frac{1}{1+x} + \frac{1}{(1+x)^2} + \cdots \\ &\sim \sum_{k=0}^{\infty} b_k \psi_k, \end{aligned}$$

where

$$b_k = 1 \qquad \forall k.$$

(III) Uniqueness of coefficients

Let $\{\phi_k\}$ be an asymptotic sequence, and

$$f(x) \sim \sum_{k=0}^{\infty} a_k \phi_k \qquad \text{as } x \to x_0.$$

Then the coefficients a_k are unique.

In fact, the a_k can be constructed:

$$a_0 = \lim_{x \to x_0} \frac{f(x)}{\phi_0(x)},$$

$$a_n = \lim_{x \to x_0} \left[\frac{f(x) - \sum_{k=0}^{n-1} a_k \phi_k(x)}{\phi_n(x)} \right].$$

Proof. First, we know

$$\lim_{x \to x_0} \frac{f(x) - \sum_{k=0}^{N} a_k \phi_k}{\phi_N} = 0 \qquad \forall N.$$

Let $N = 0$. Then

$$\lim_{x \to x_0} \frac{f(x) - a_0 \phi_0}{\phi_0} = 0,$$

$$\lim_{x \to x_0} \left(\frac{f(x)}{\phi_0} \right) - a_0 = 0.$$

Use induction on N.

Suppose the formulae are true for $N = n$, and the coefficients are unique for $k \leq n$. Then

$$\lim_{x \to x_0} \frac{f(x) - \sum_{k=0}^{n} a_k \phi_k}{\phi_n} = 0$$

$$= \lim_{x \to x_0} \left(\frac{f(x) - \sum_{k=0}^{n-1} a_k \phi_k}{\phi_n} \right) - a_n.$$

Alternative proof: see p. 3 in Manton.

(IV) An asymptotic series does not uniquely determine the function it is describing

For example, suppose

$$f(x) \sim g(x) \qquad \text{as } x \to x_0,$$

and

$$h(x) \ll f(x) \qquad \text{as } x \to x_0.$$

Then

$$f(x) + h(x) \sim g(x) \qquad \text{as } x \to x_0.$$

So, if g is an asymptotic expansion, it can describe both f and $f+h$.

For example,

$$f_1 = \frac{1}{1+x}, \qquad f_2 = \frac{1+e^{-x}}{1+x}.$$

Both are

$$\sim \sum_{k=0}^{\infty} (-1)^k x^{-(k+1)} \qquad \text{as } x \to \infty.$$

(V) Power series

In this course, we mostly take

$$\phi_k = (x - x_0)^k \qquad \text{as } x \to x_0,$$

or

$$\phi_k = x^{-k} \qquad \text{as } x \to \infty.$$

In fact, a Taylor series about a point is an asymptotic series about that point:

$$f(x) - \sum_{k=0}^{N} \frac{f^{(k)}(a)(x-a)^k}{k!} = o\big(|x-a|^N\big)\,.$$

Exercise: Prove this using the definitions and the integral form of the remainder term.

Terminology: A term is said to be "small beyond all orders" in some limit if it cannot be represented by any power series or, equivalently, is smaller than any power.

For example, e^{-x} is small beyond all orders in inverse powers of x as $x \to \infty$:

$$e^{-x} \ll x^{-k} \qquad \forall k, \qquad \text{as } x \to \infty.$$

(VI) Optimal truncation

Question: If an asymptotic series diverges, how many terms should I include in the sum?

If

$$f(x) \sim \sum_{n=0}^{\infty} a_n \psi_n(x) \qquad \text{as } x \to x_0,$$

and we want to get the best approximation to $f(x)$ at $x = x_1$, where x_1 is close to x_0, then truncate the sum at the smallest term (or the one before).

So stop adding terms when

$$a_k \psi_k(x_1) < a_{k+1} \psi_{k+1}(x_1).$$

This gives you an unexpectedly excellent approximation.

End of Lecture 2

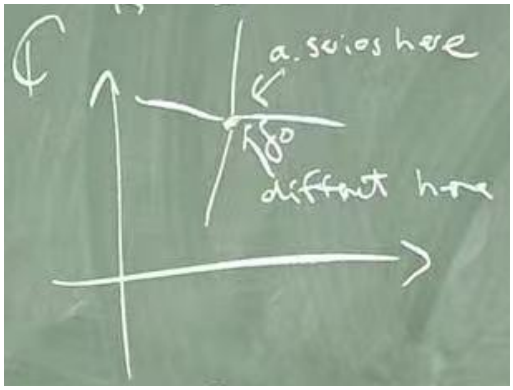
2.3 A. expansions in $\mathbb{C}$ — Stokes phenomenon

- So far our defns relied on real limits, $x, x_0 \in \mathbb{R}$ etc.
- When we use complex variables, $x = z \in \mathbb{C}$ and $z \to z_0$, typically z must be restricted to a sector of $\mathbb{C}$ around z_0 for some of our definitions to work.

  [See III.2 in Stuart]

  e.g. little o & O defns need to bring in moduli of functions.

- Even if $f(z)$ analytic at z_0, it may have different a. series as z_0 is approached from different sectors in $\mathbb{C}$. This is Stokes phenomenon.



e.g.

$$f(z) = \sinh(z^2) = \frac{1}{2}\left(e^{z^2} - e^{-z^2}\right), \qquad z \in \mathbb{C}.$$

What is its behaviour as $|z| \to \infty$?

Which of e^{z^2} or e^{-z^2} is dominant?

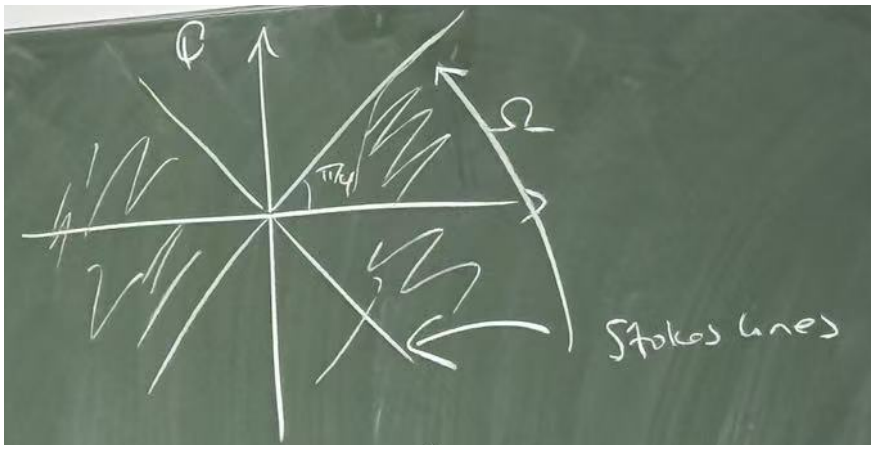
$$\left|\frac{e^{-z^2}}{e^{z^2}}\right| = \left|e^{-2z^2}\right| = e^{-2\,\mathrm{Re}(z^2)}.$$

(a) Suppose $|z| \to \infty$ & $\mathrm{Re}(z^2) > 0$.

$$\text{ratio} \to 0 \quad \Rightarrow \quad e^{-z^2} \ll e^{z^2} \quad \Rightarrow \quad f(z) \sim \frac{1}{2}e^{z^2} \quad \text{in this limit.}$$

$$\mathrm{Re}(z^2) > 0 \quad \Rightarrow \quad \Omega: \quad -\frac{\pi}{4} + \delta < \arg z < \frac{\pi}{4} - \delta, \quad \text{and} \quad \frac{3\pi}{4} + \delta < \arg z < \frac{5\pi}{4} - \delta,$$

for small δ.



(b) $|z| \to \infty$ & $\mathrm{Re}(z^2) < 0$.

$$\text{ratio} \to \infty \quad \Rightarrow \quad e^{z^2} \ll e^{-z^2} \quad \& \quad f \sim -\frac{1}{2}e^{-z^2}.$$

$$\widetilde{\Omega}: \quad \frac{\pi}{4}+\delta < \arg z < \frac{3\pi}{4}-\delta \quad \& \quad -\frac{3\pi}{4}+\delta < \arg z < -\frac{\pi}{4}-\delta.$$

$$f(z) \sim \begin{cases} \frac{1}{2}e^{z^2}, & |z|\to\infty \ \& \ z\in\Omega, \\ -\frac{1}{2}e^{-z^2}, & |z|\to\infty \ \& \ z\in\widetilde{\Omega}. \end{cases}$$

- A. behaviour of f as $|z|\to\infty$ depends on which wedge $z\to\infty$ within.
- $\mathbb{C}$ is divided into sectors, separated by special lines, called Stokes lines. [Warning, sometimes called anti-Stokes lines!!]

3. Approximations of Integrals

3.1 Taylor expanding the integrand — Stieltjes integrals

(Warm up problem)

Consider:

$$I(x) = \int_0^\infty \frac{e^{-t}}{1+xt}\,dt, \qquad x\geq 0.$$

- We want to understand I's behaviour when $x\to 0^+$.
- How about we expand $\dfrac{1}{1+xt}$?

$$\frac{1}{1+xt} = \sum_{n=0}^{\infty}(-1)^n x^n t^n \qquad \times \quad t \text{ can be big!}$$

Consider instead

$$\sum_{n=0}^{N}(-xt)^n = \frac{1-(-xt)^{N+1}}{1+xt} = \frac{1}{1+xt} - \underbrace{\frac{(-xt)^{N+1}}{1+xt}}_{r_N}.$$

$$\Rightarrow \quad \frac{1}{1+xt} = \sum_{n=0}^{N}(-xt)^n + r_N.$$

$$\begin{aligned} I(x) &= \int_0^\infty e^{-t}\left[\sum_{n=0}^{N}(-1)^n x^n t^n + r_N\right]dt \\ &= \sum_{n=0}^{N}(-1)^n x^n \underbrace{\int_0^\infty t^n e^{-t}\,dt}_{=n!} + R_N \\ &= \sum_{n=0}^{N}(-1)^n n!\, x^n + R_N \\ &= \sum c_n \phi_n(x) + R_N. \end{aligned}$$

$\phi_n = x^n$ as $x\to 0^+$, so is a. sequence!

Need to show $R_N = o(x^N)$ as $x\to 0^+$.

$$\begin{aligned} \frac{R_N}{x^N} &= \frac{x^{N+1}}{x^N}\int_0^\infty \frac{t^{N+1}e^{-t}}{1+xt}\,dt \\ &\leq x\int_0^\infty t^{N+1}e^{-t}\,dt = x(N+1)! \longrightarrow 0 \quad \text{as } x\to 0^+. \end{aligned}$$

Note $1 + xt > 1$. True $\forall N$.

$$\Rightarrow \quad I(x) \sim \sum_{n=0}^{\infty} (-1)^n n!\, x^n \qquad \text{as } x \to 0^+.$$

3.2 Integration by parts

Recap:

$$\int_a^b f'(t)g(t)\,dt = [fg]_a^b - \int_a^b fg'\,dt.$$

Generalisation:

$$\begin{aligned} I &= \int_a^b f(t)g^{(n)}(t)\,dt \\ &= \sum_{k=1}^{n} (-1)^{k-1} \left[f^{(k-1)}(t)g^{(n-k)}(t) \right]_a^b + (-1)^n \int_a^b f^{(n)}(t)g(t)\,dt. \end{aligned}$$

- Repeated integration by parts can generate an a. approximation under certain circumstances.

e.g. How does

$$F(x) = \int_x^\infty e^{-t^2/2}\,dt$$

behave as $x \to \infty$?

$$\begin{aligned} F(x) &= \int_x^\infty \frac{te^{-t^2/2}}{t}\,dt \\ &= \left[-\frac{1}{t} e^{-t^2/2} \right]_x^\infty + \int_x^\infty \frac{e^{-t^2/2}}{t^2}\,dt \\ &= \underbrace{\frac{e^{-x^2/2}}{x}}_{\text{a. approximation}} + \underbrace{\int_x^\infty \frac{e^{-t^2/2}}{t^2}\,dt}_{\text{error}}. \end{aligned}$$

Need to check that

$$\text{error} = o\left(\frac{e^{-x^2/2}}{x} \right) \qquad \text{as } x \to \infty.$$

$$\begin{aligned} \text{error} &= \int_x^\infty \frac{te^{-t^2/2}}{t^3}\,dt \qquad (\text{note } t \geq x) \\ &\leq \frac{1}{x^3} \int_x^\infty te^{-t^2/2}\,dt = \frac{1}{x^3} e^{-x^2/2} \\ &= o\left(\frac{e^{-x^2/2}}{x} \right) \qquad \text{as } x \to \infty. \end{aligned}$$

$$\Rightarrow \quad F(x) \sim \frac{e^{-x^2/2}}{x} \qquad \text{as } x \to \infty.$$

In fact we find

$$F(x) \sim \frac{e^{-x^2/2}}{x} \sum_{n=0}^{\infty} (-1)^n \frac{(2n-1)!!}{x^{2n}} \qquad \text{as } x \to \infty.$$

$$(2n-1)!! = (2n-1)(2n-3)(2n-5)\cdots 3 \cdot 1.$$

Exercise: Show this.

End of Lecture 3